# SEG2105 - Assignment 2

## Requirements and Object-Oriented Modelling

## OConnect City Transit Management System

**Nathan Ng** - 300494064
**Travis Zhang** - 300488899

Fall 2026

Answers are presented in question order. Large UML diagrams are divided into labelled views of one model; repeated object or class names identify the same element across views. All diagrams are embedded as high-resolution images.

## Contents

Question 1 - UML class diagram and constraints

Question 2 - UML object diagram and scenario assumptions

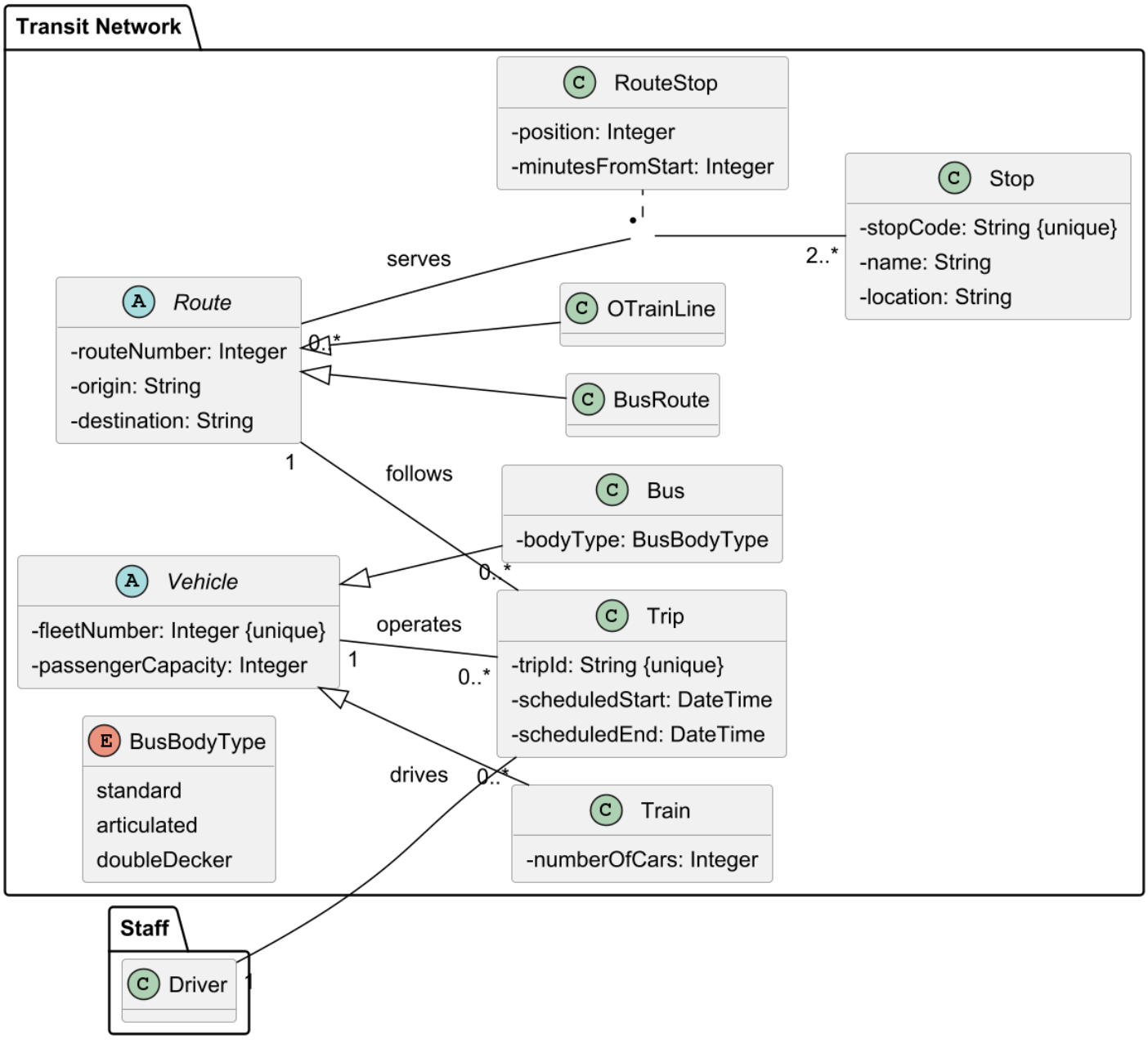
Question 3 - Five functional requirements

Question 4 - Nine non-functional requirements

Question 5 - UML use case diagram

Question 6 - Detailed use case: Pay fare

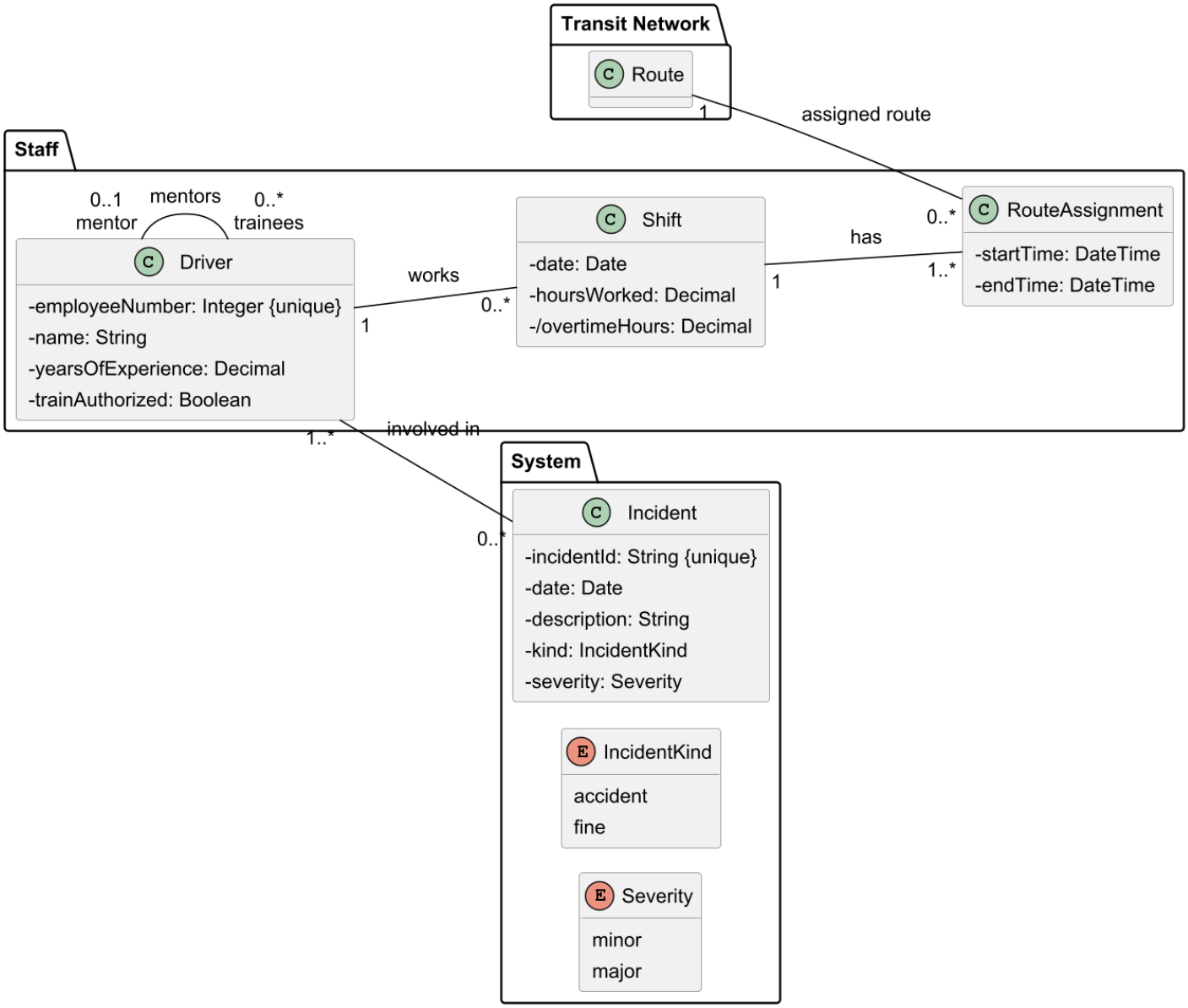
# Question 1 - Domain model: network and fleet




RouteStop is a UML association class: its dashed connector attaches it to the Route-Stop association. Position and travel time therefore describe a stop on a particular route. Driver attributes appear in the staff view. Generalization sets for Route and Vehicle are complete and disjoint.

**Assumptions:** A route is one direction with at least two distinct stops and no repeated stop. Origin and destination name its first and last stops. Route numbers are unique within each route subtype. All Trip objects represent fully allocated scheduled runs.

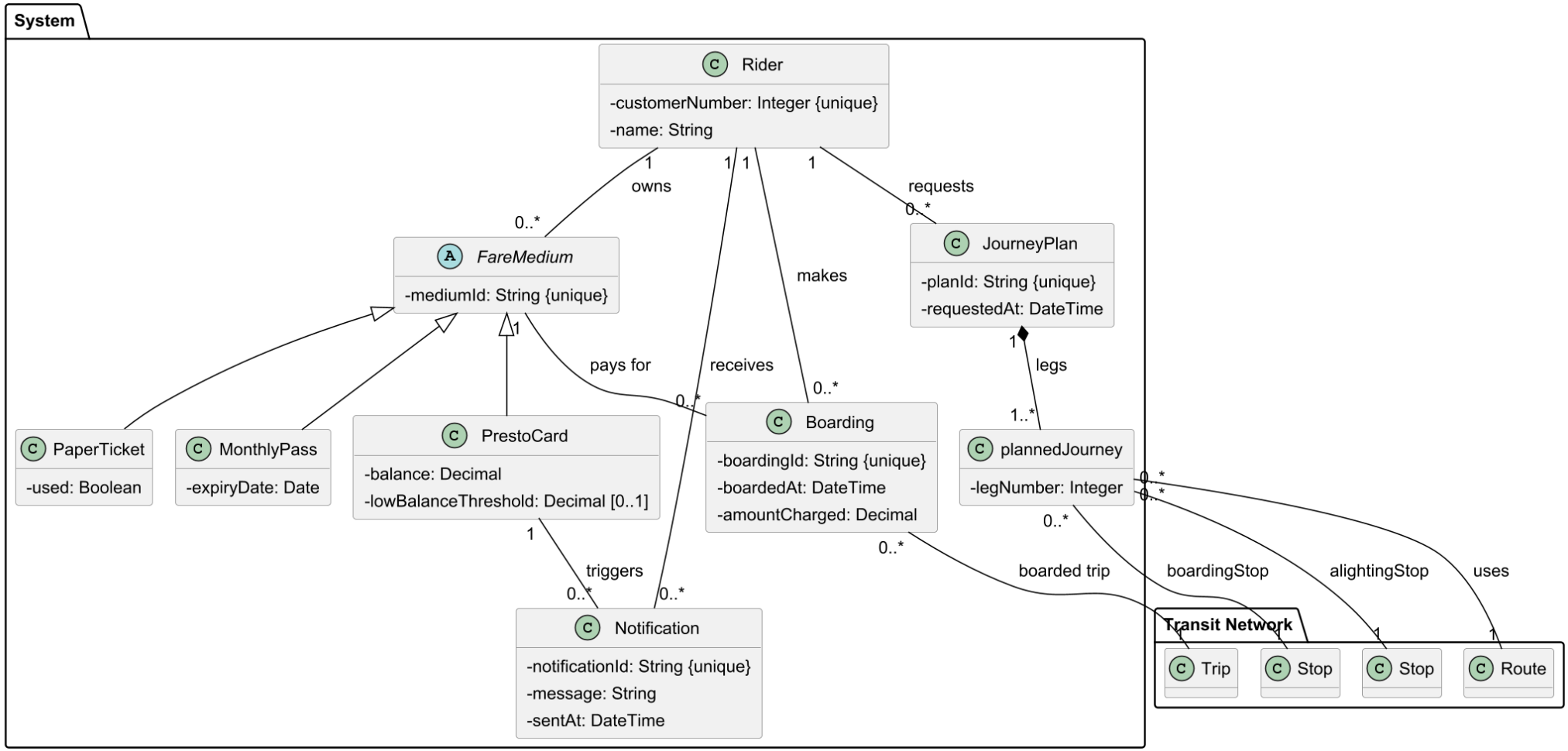
# Question 1 - Domain model: staff




**Constraints:** A driver with experience < 2 years has exactly one mentor; all others have none. A mentor has at least 2 years of experience and cannot mentor themself. Train authorization implies experience >= 10 years; experience alone does not grant authorization.

**Assumptions:** One Shift per driver per date. Hours are decimal paid working hours. /overtimeHours = max(0, hoursWorked - 7.5). Assignment intervals are within that shift, do not overlap, and cover its working time. Each incident can involve multiple drivers; its date, kind and severity are shared. Accident/fine and minor/major are enumeration values, not subclasses.

# Question 1 - Domain model: riders and fares




**Assumptions:** Fare media are registered to one rider and are not shared. FareMedium specializations are complete and disjoint. Boarding records successful payments only. A JourneyPlan records a successful itinerary with one or more ordered legs; it is distinct from a scheduled Trip. Stops and routes are shared network entities. Notification records represent messages sent to a rider.

# Question 1 - Types, constraints and interpretation

**Types.** String is text; Integer is a whole number; Decimal is an exact fixed-point number; Boolean is true/false; Date and DateTime use city local time. Monetary values are CAD to two decimal places. A type followed by [0..1] is optional. A leading slash denotes a derived attribute.

**RouteStop.** For each route, positions are unique and consecutive from 1. The first stop has minutesFromStart = 0; later offsets are nonnegative and nondecreasing. A Route-Stop pair occurs once. The association class is equivalent to an ordered route membership record, not an attribute of Stop.

**Trips and assignments.** A BusRoute trip uses a Bus; an OTrainLine trip uses a Train and a train-authorized driver. An unauthorized driver is assigned only bus routes. Authorized drivers may also drive buses. Each trip lies within a RouteAssignment for the same route and driver (through Shift). A driver or vehicle cannot operate overlapping trips.

**Numerical constraints.** Capacity and numberOfCars are positive; yearsOfExperience, hoursWorked, balance and any threshold are nonnegative. Scheduled start precedes scheduled end. Assignment start precedes assignment end.

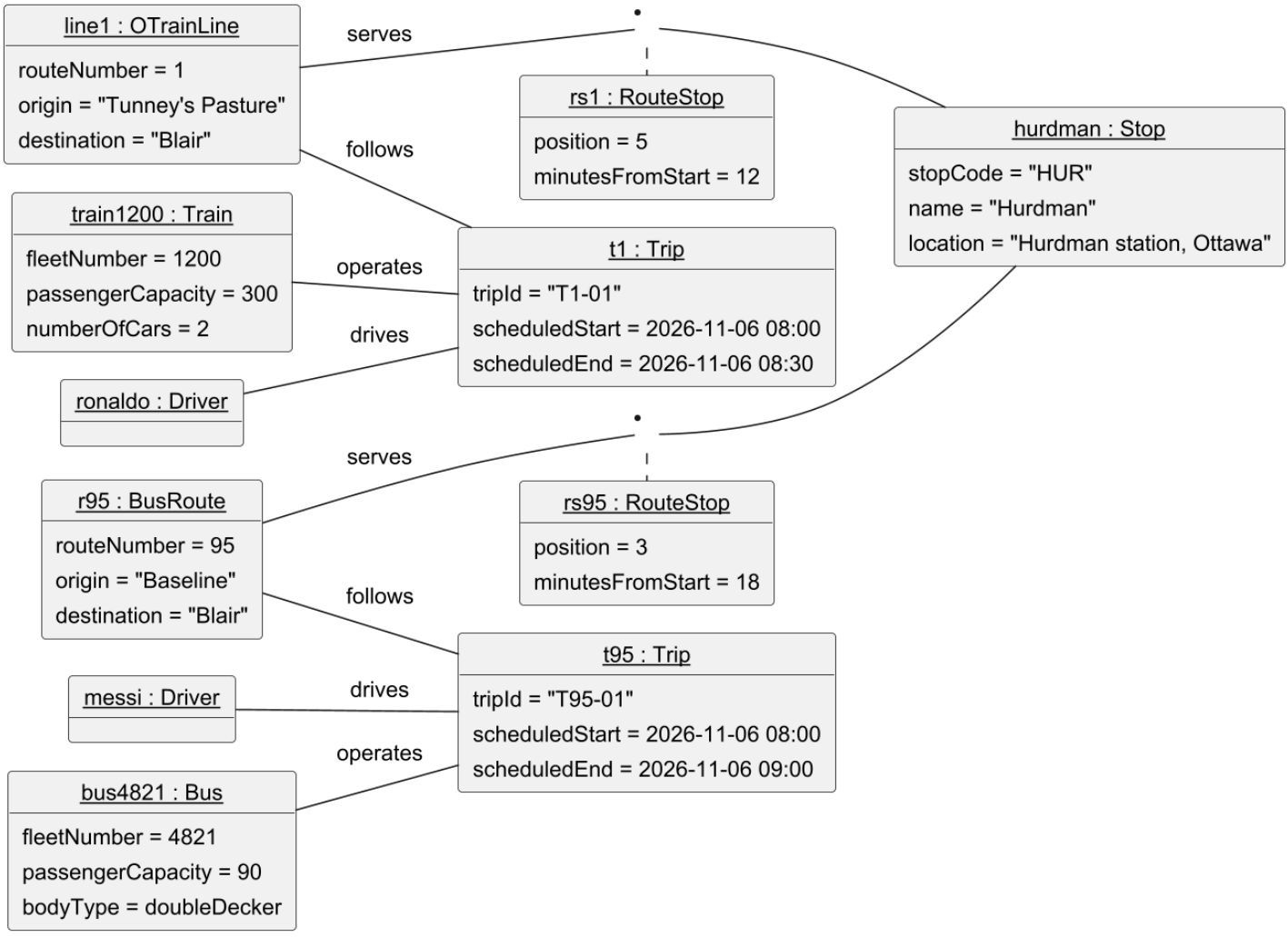
**Fare rules.** A PaperTicket pays for at most one Boarding, and used is true exactly when that boarding exists. A MonthlyPass permits unlimited boardings through the end of expiryDate. A PrestoCard payment cannot make its balance negative. Boarding.amountCharged is the actual card debit, or 0 for a ticket/pass whose purchase occurred earlier.

**Ownership and notifications.** A Boarding rider owns the fare medium used. A Notification recipient owns the triggering PrestoCard. After a successful card payment, a configured threshold greater than the new balance triggers one notification for that payment; no threshold means no alert.

**Journey plans.** plannedJourney numbers are consecutive from 1. Each leg uses two stops served by its route, in increasing RouteStop position. Adjacent legs meet at the same transfer stop. The first boardingStop and final alightingStop identify the requested origin and destination.

**Model scope.** RouteAssignment, Boarding and plannedJourney are ordinary classes with binary associations, not UML association classes. RouteStop is the only association class. OConnect itself is the system boundary, not an unnecessary all-owning domain class. Operations are omitted because this is a structural domain model.

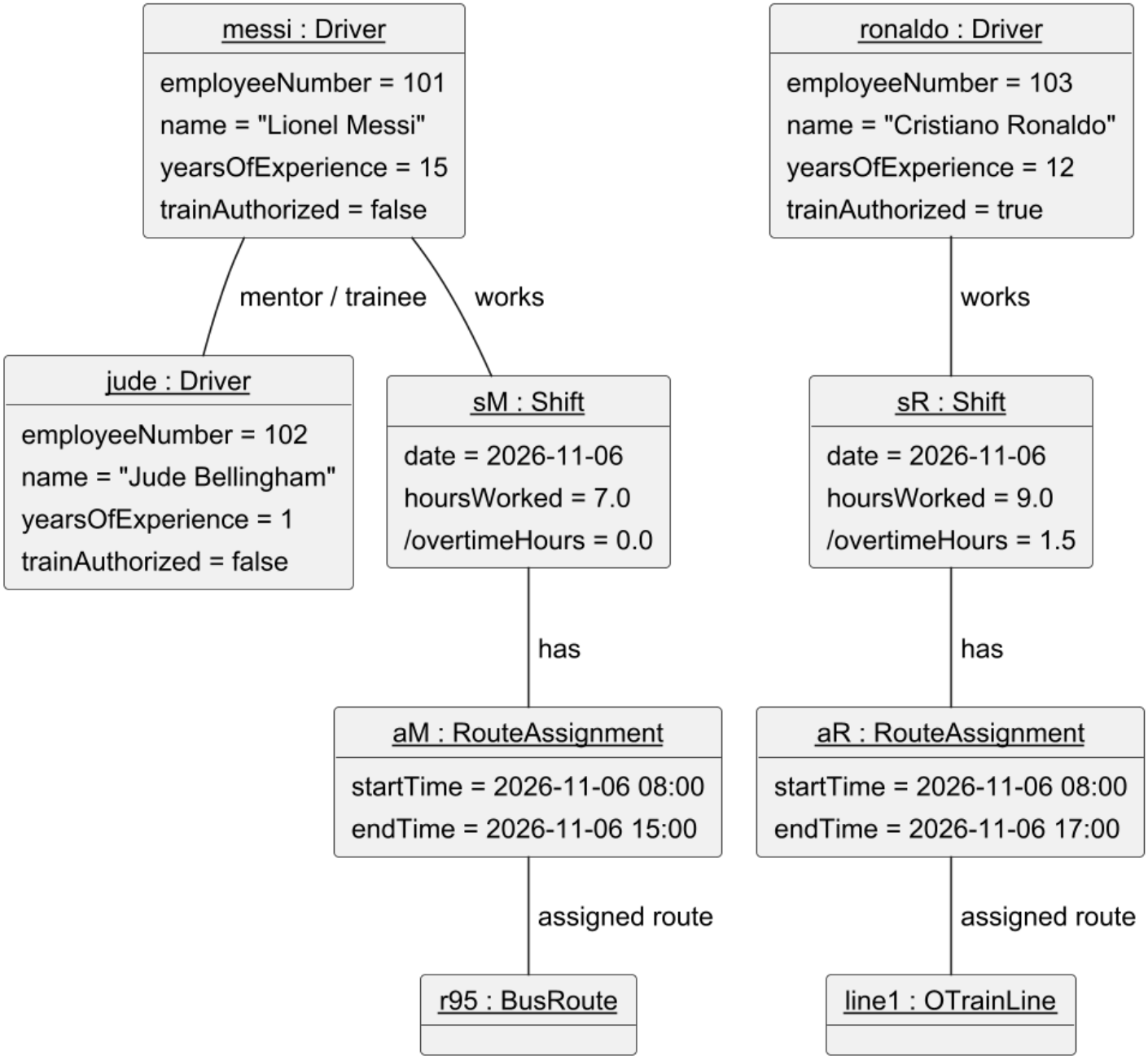
# Question 2 - Object diagram: routes and trips




**Snapshot assumptions:** The day is 2026-11-06. Capacities, car count, identifiers and times are illustrative. This is a partial network snapshot: other stops, RouteStop links and daily trips are omitted. Hurdman is the same Stop in both associations. The two dashed association objects instantiate the sole association class in Question 1.

Underlined object headers use objectName : Class. Repeated object names across the following views denote the same instances; empty repeated boxes are references, not objects lacking values.

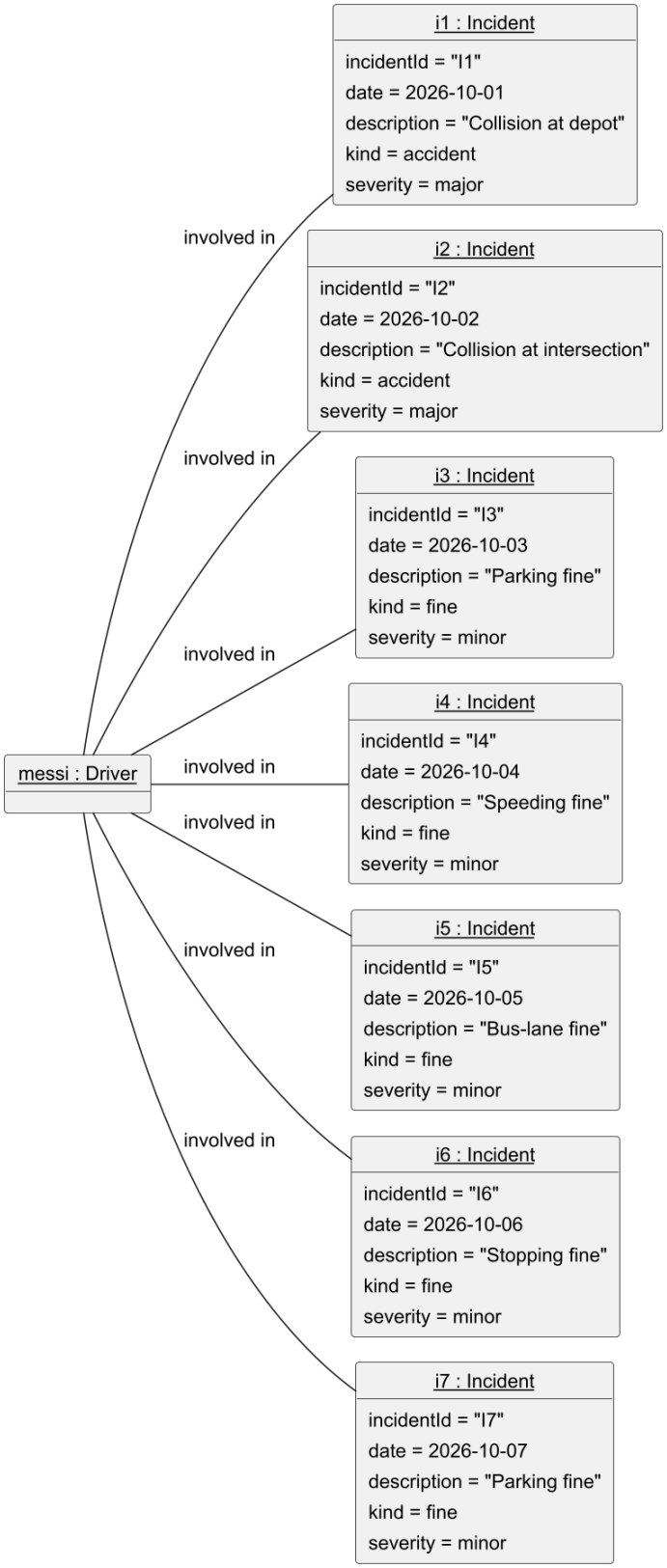
# Question 2 - Object diagram: drivers and shifts




**Assumptions (Questions 1-2 only):** Messi is assigned 15 years of experience and trainAuthorized = false because neither value is specified. Employee numbers 101-103 are illustrative. Messi and Ronaldo each keep one route assignment for their entire recorded shift, with no unpaid break. Their shifts start at 08:00. No shift for Jude is shown because Question 2 does not specify his work that day.

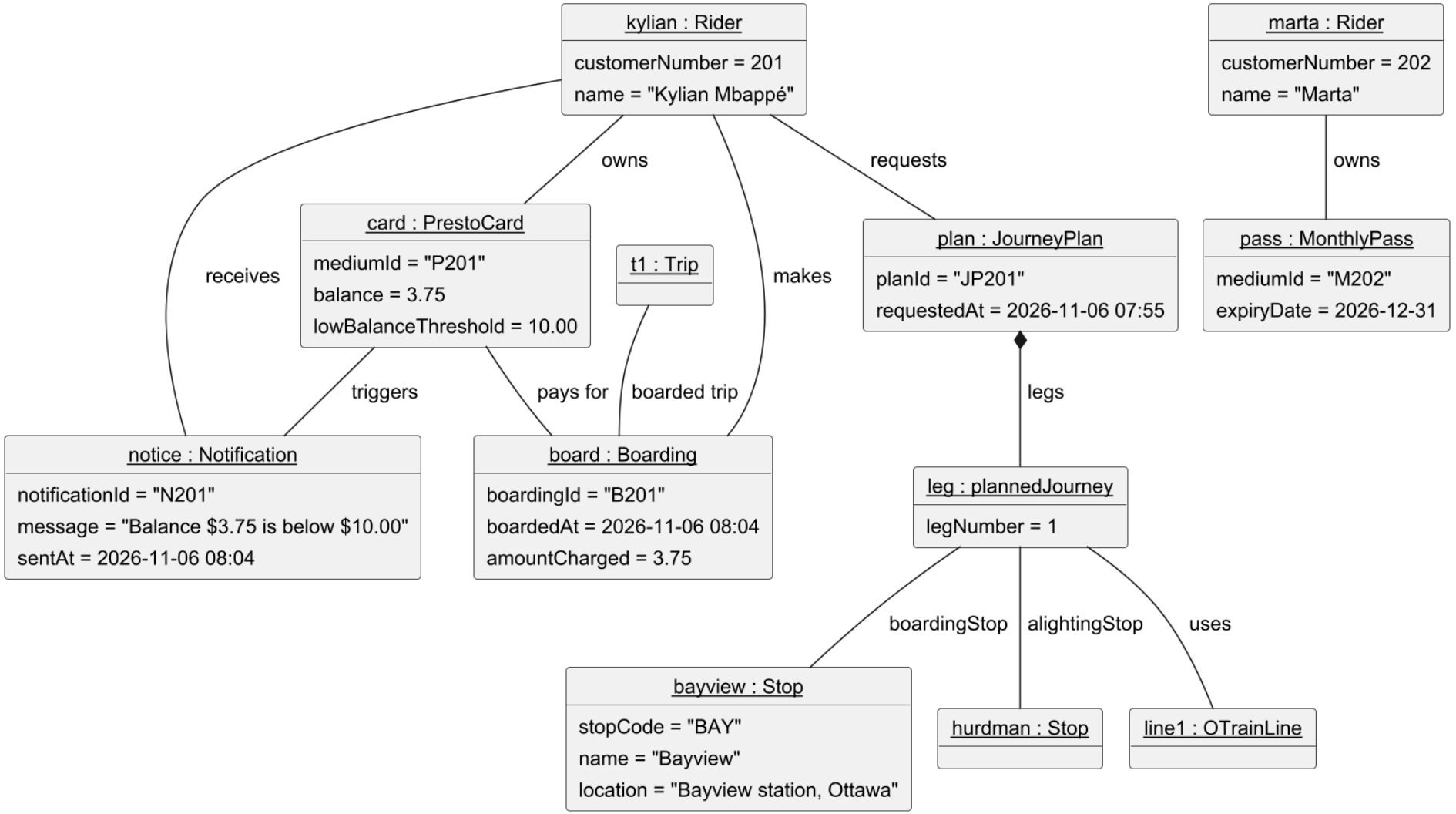
**Given facts and consequences:** Question 2 identifies Messi as Jude's mentor and gives Ronaldo 12 years of experience, a train assignment and no incidents. Ronaldo therefore has train authorization under the Question 1 model. Using the stated 7.5-hour daily rule, Messi's 7 hours give 0 overtime and Ronaldo's 9 hours give 1.5 hours.

# Question 2 - Object diagram: Messi's incidents




**Assumptions:** Dates and short descriptions are illustrative; all five fines are assumed minor because their severity was not specified. These are Messi's complete incident links: exactly two major accidents and five fines. Ronaldo has none.

# Question 2 - Object diagram: riders and fares




**Assumptions:** The $3.75 balance is after Kylian's payment of an assumed $3.75 fare from $7.50. He planned the Bayview-Hurdman journey before boarding t1 at Bayview. Bayview is assumed to be Line 1 position 2, offset 4 minutes (its RouteStop link is omitted). Hurdman is position 5, offset 12. The alert has been sent. Marta owns the shown pass; her particular boarding is unspecified.

# Question 3 - Functional requirements

Each requirement covers a different part of OConnect. The proposed rules use the domain assumptions in Question 1.

**FR1.** The system shall show routes and transfer stops connecting a rider's chosen starting stop and destination when a connection exists.

**FR2.** The system shall save each confirmed route assignment with the driver, route, start time and end time.

**FR3.** The system shall count any time a driver works beyond 7.5 hours in a day as overtime.

**FR4.** The system shall record each driver incident with the drivers involved, date, description, type and severity.

**FR5.** The system shall notify a rider after a successful Presto payment leaves their card balance below their chosen threshold.

**Assumption:** FR1 addresses route connectivity, without live service disruptions or a fastest-arrival guarantee. FR5 uses a strict less-than comparison; a balance equal to the threshold does not trigger an alert.

# Question 4 - Non-functional requirements

The following are proposed acceptance constraints, because the scenario does not specify deployment or quality targets. Release acceptance uses the stated workloads and environments.

## Quality (5)

**NFR1 - Performance.** With 500 concurrent active riders and a network of 200 routes and 2,000 stops, at least 95% of trip-planning requests shall return a result within 2 seconds over a 30-minute acceptance run, measured from server receipt to response completion.

**NFR2 - Availability.** The fare-validation service shall achieve at least 99.9% availability in each calendar month, measured by a successful synthetic fare-validation request every minute, with planned downtime included as unavailable.

**NFR3 - Usability.** At least 18 of 20 first-time rider participants shall plan a direct trip between two named stops within 2 minutes without assistance during a moderated acceptance test.

**NFR4 - Reliability.** In a 10,000-request replay test in which each successful Presto payment request is resent twice with the same transaction identifier, the persisted balance shall reflect exactly one fare deduction per distinct transaction identifier.

**NFR5 - Recoverability.** After simulated loss of the primary database, the operations team shall restore the service to its latest recoverable state within 30 minutes using the documented recovery procedure in a quarterly recovery test.

## Platform (2)

**NFR6.** The rider web interface shall pass its complete core-workflow acceptance suite on the latest two stable major versions of Chrome, Firefox, Edge and Safari available on the release freeze date.

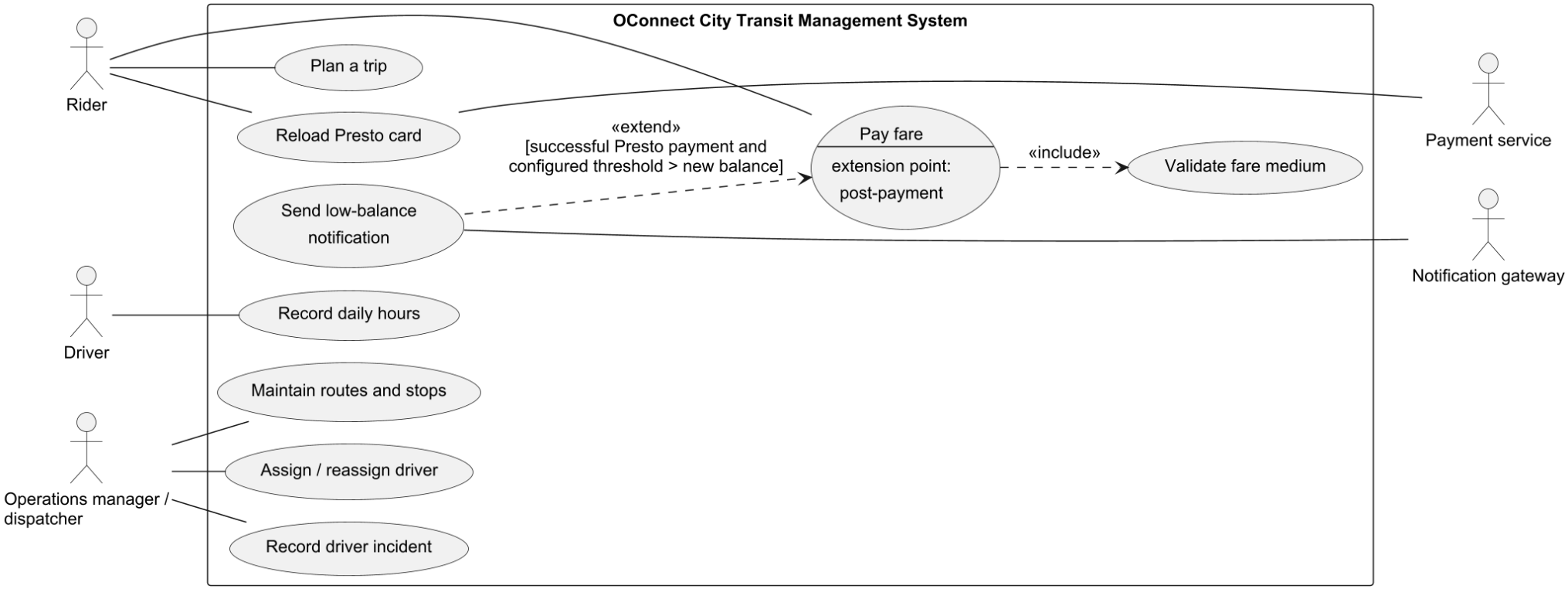
**NFR7.** The application server shall pass its complete integration suite in a Linux x86-64 container limited to 4 virtual CPUs and 8 GiB of RAM.

## Process (2)

**NFR8.** Every production code change shall have a recorded approval from at least one developer other than its author before it is merged into the release branch.

**NFR9.** Each release shall have an archived acceptance-test report linking every approved functional requirement to at least one passing test before deployment approval.

# Question 5 - UML use case diagram




**Relationships:** Pay fare always includes Validate fare medium. Send low-balance notification extends Pay fare at post-payment only under the shown condition; the dashed arrow points to the base use case. Reload Presto card uses the external payment service.

**Assumptions:** Riders interact through a website or onboard reader; these are interfaces within the system, not independent external actors. Drivers record hours; dispatchers manage assignments, incidents and routes. Card reloads use an external payment service, while alerts use an external notification gateway. Primary actors are on the left, external actors on the right. No actor or use case generalization is needed: no genuine substitutable specialization is introduced.

# Question 6 - Detailed use case: Pay fare

**Identifier and scope.** UC-PAY; OConnect City Transit Management System; user-goal level.

**Primary actor.** Rider.

**Supporting actor.** Notification gateway, when the low-balance extension applies.

**Stakeholders.** The rider wants a valid fare accepted and an accurate balance; OConnect wants authorized boarding and accurate fare records; the dispatcher wants a clear acceptance or rejection result.

**Trigger.** The rider presents a paper ticket, monthly pass or Presto card at the onboard reader for a scheduled trip.

**Preconditions.** The trip is active and has a route, vehicle and driver. The reader identifies the trip and can contact OConnect. The rider and presented fare medium are registered. Applicable fare rules are configured.

**Success guarantee.** Exactly one Boarding is recorded for the transaction and boarding is authorized. A Presto balance is debited once, a ticket is marked used, or a pass is accepted without per-ride debit. If a low-balance alert is required, it is sent or queued for retry.

**Minimum guarantee.** A rejected or uncommitted payment does not consume the ticket, debit the card or create a successful Boarding. Retrying a committed transaction cannot charge it again.

## Main success flow (Presto card)

1. The rider taps their Presto card at the trip's reader.

2. The reader submits the medium identifier, trip identifier and a unique transaction identifier.

3. OConnect invokes **Validate fare medium**, identifies the owner, and verifies that the card is recognized and usable.

4. OConnect determines the applicable fare and verifies that the available balance covers it.

5. OConnect atomically deducts the fare and records the successful Boarding with the rider, medium, trip, timestamp and amount.

6. At extension point **post-payment**, if a configured threshold is greater than the new balance, OConnect invokes **Send low-balance notification**.

7. OConnect returns acceptance and the new balance to the reader; the rider boards.

# Question 6 - Alternatives and exceptions

**3a - Paper ticket.** Validate fare medium confirms the ticket is unused. At step 5, OConnect atomically marks it used and records a Boarding with amountCharged = 0. Step 6 does not apply; continue at step 7. An already-used ticket is rejected without state changes.

**3b - Monthly pass.** Validation checks that the local boarding date is on or before expiryDate. At step 5, OConnect records a Boarding with amountCharged = 0 without changing the expiry date. Skip step 6 and continue at step 7. An expired pass is rejected.

**3c - Unrecognized medium.** OConnect displays a rejection reason and records no successful Boarding. The rider may restart with another medium.

**4a - Insufficient balance.** OConnect rejects the payment and displays the shortfall without debiting the card. The rider may complete the separate Reload Presto card use case or select another medium, then restart Pay fare.

**5a - Failure before commit.** OConnect rolls back the transaction and the reader does not authorize boarding. A retry uses the same transaction identifier; if no commit occurred, processing resumes normally.

**5b - Duplicate request or lost reply.** If the transaction identifier already identifies a committed Boarding, OConnect returns its recorded acceptance result without a second debit, ticket consumption, Boarding or low-balance notification. The Boarding identifier serves as the transaction identifier.

**6a - No alert required.** If the threshold is absent or the balance is greater than or equal to it, OConnect skips the extension and continues at step 7.

**6b - Notification delivery failure.** OConnect retains a retry task associated with the transaction and continues at step 7. Failure to send an alert does not reverse an accepted fare. A Notification domain record is created when dispatch succeeds; transport retry tasks are implementation details outside the domain diagram.

**Special requirements.** NFR2 governs availability; NFR4 governs transaction replay. Fare updates and Boarding creation are atomic. Money uses exact decimal arithmetic.

**Assumptions and template.** A ticket is consumed for one boarding on one scheduled trip; transfer privileges are outside scope. A pass is valid through 23:59:59 local time on its expiry date. Offline acceptance is outside scope. Since no course use-case template was supplied, this description uses the standard fields: scope, level, actors, stakeholders, trigger, preconditions, guarantees, main flow, alternatives and special requirements.